Series : A4BAB/3

SET-1

प्रश्न–पत्र कोड
Q.P. Code **55/3/1**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न–पत्र कोड को उत्तर–पुस्तिका के मुख–पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

नोट

(I) कृपया जाँच कर लें कि इस प्रश्न–पत्र में मुद्रित पृष्ठ **12** हैं ।

(II) प्रश्न–पत्र में दाहिने हाथ की ओर दिए गए प्रश्न–पत्र कोड को छात्र उत्तर–पुस्तिका के मुख–पृष्ठ पर लिखें ।

(III) कृपया जाँच कर लें कि इस प्रश्न–पत्र में **12** प्रश्न हैं ।

(IV) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, प्रश्न का क्रमांक अवश्य लिखें ।**

(V) इस प्रश्न–पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न–पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न–पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर–पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।

NOTE

(I) Please check that this question paper contains **12** printed pages.

(II) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(III) Please check that this question paper contains **12** questions.

(IV) **Please write down the Serial Number of the question in the answer-book before attempting it.**

(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period. *

भौतिक विज्ञान (सैद्धान्तिक)

PHYSICS (Theory)

निर्धारित समय : 2 घण्टे — *अधिकतम अंक : 35*

Time allowed : 2 hours — *Maximum Marks : 35*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains 12 questions. **All** questions are compulsory.*

(ii) *This question paper is divided into three sections – Section A, B and C.*

(iii) ***Section A :** Q. Nos. **1** to **3** are of **2** marks each.*

(iv) ***Section B :** Q. Nos. **4** to **11** are of **3** marks each.*

(v) ***Section C :** Q. No. **12** is a case study based questions of **5** marks.*

(vi) *There is no overall choice in the question paper. However, internal choice has been provided in some of the questions. Attempt any one of the alternatives in such questions.*

(vii) *Use of log tables is permitted, if necessary, but use of calculator is not permitted.*

$c = 3 \times 10^8$ m/s

$h = 6.63 \times 10^{-34}$ Js

$e = 1.6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2 N^{-1} m^{-2}

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N m^2 C^{-2}

Mass of electron (m_e) = 9.1×10^{-31} kg

Mass of neutron = 1.675×10^{-27} kg

Mass of proton = 1.673×10^{-27} kg

Avogadro's number = 6.023×10^{23} per gram mole

Boltzmann constant = 1.38×10^{-23} JK^{-1}

SECTION-A

1. What is meant by doping of an intrinsic semiconductor ? Name the two types of atoms used for doping of Ge/Si. **2**

2. (a) (i) Distinguish between isotopes and isobars. **2**

   (ii) Two nuclei have different mass numbers A_1 and A_2. Are these nuclei necessarily the isotopes of the same element ? Explain.

   OR

   (b) (i) Name the factors on which photoelectric emission from a surface depends.

   (ii) Define the term 'threshold frequency' for a photosensitive material.

3. Explain the formation of the barrier potential in a p-n junction. **2**

SECTION-B

4. State Bohr's postulate to explain stable orbits in a hydrogen atom. Prove that the speed with which the electron revolves in n^{th} orbit is proportional to (1/n). **3**

5. Briefly explain how emf is generated in a solar cell. Draw its I-V characteristics. **3**

6. A narrow beam of protons, each having 4.1 MeV energy is approaching a sheet of lead (Z = 82). Calculate :

   (i) the speed of a proton in the beam, and

   (ii) the distance of its closest approach **3**

7. In a diffraction pattern due to a single slit, how will the angular width of central maximum change, if

   (i) Orange light is used in place of green light,

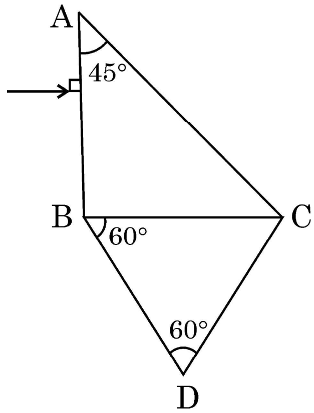
   (ii) the screen is moved closer to the slit,

   (iii) the slit width is decreased ?

   Justify your answer in each case. **3**

8. (a) Write two necessary conditions for total internal reflection.

(b) Two prisms ABC and DBC are arranged as shown in figure.

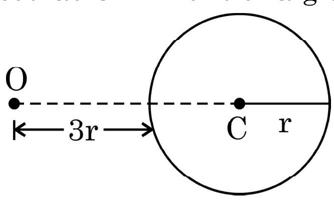


The critical angles for the two prisms with respect to air are 41.1° and 45° respectively. Trace the path of the ray through the combination. **3**

OR

(a) An object is placed in front of a converging lens. Obtain the conditions under which the magnification produced by the lens is (i) negative and (ii) positive.

(b) A point object is placed at O in front of a glass sphere as shown in figure.



Show the formation of image by the sphere.

9. An electron is accelerated from rest through a potential difference of 100 V. Find :

(i) the wavelength associated with

(ii) the momentum of and

(iii) the velocity required by, the electron. **3**

10. In a Young's double slit experiment using light of wavelength 600 nm, the slit separation is 0.8 mm and the screen is kept 1.6 m from the plane of the slits. Calculate :

(i) the fringe width

(ii) the distance of (a) third minimum and (b) fifth maximum, from the central maximum. **3**

11. (a) Electromagnetic waves of wavelengths λ_1, λ_2 and λ_3 are used in radar systems, in water purifiers and in remote switches of TV, respectively.

(i) Identify the electromagnetic waves, and

(ii) Write one source of each of them. **3**

OR

(b) (i) State two conditions for two light sources to be coherent.

(ii) Give two points of difference between an interference pattern due to a double – slit and a diffraction pattern due to a single slit.

SECTION-C
CASE STUDY

12. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image. **1 × 5 = 5**

I. In a compound microscope the images formed by the objective and the eye-piece are respectively

(A) virtual, real (B) real, virtual

(C) virtual, virtual (D) real, real

II. The magnification due to a compound microscope *does not* depend upon

(A) the aperture of the objective and the eye-piece

(B) the focal length of the objective and the eye-piece

(C) the length of the tube

(D) the colour of the light used

III. Which of the following is *not correct* in the context of a compound microscope ?

(A) Both the lenses are of short focal lengths.

(B) The magnifying power increases by decreasing the focal lengths of the two lenses.

(C) The distance between the two lenses is more than (fo + fe).

(D) The microscope can be used as a telescope by interchanging the two lenses.

IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2 (B) 10

(C) 30 (D) 200

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100 (B) 150

(C) 200 (D) 250